

CSE-251

ELECTRONIC CIRCUITS

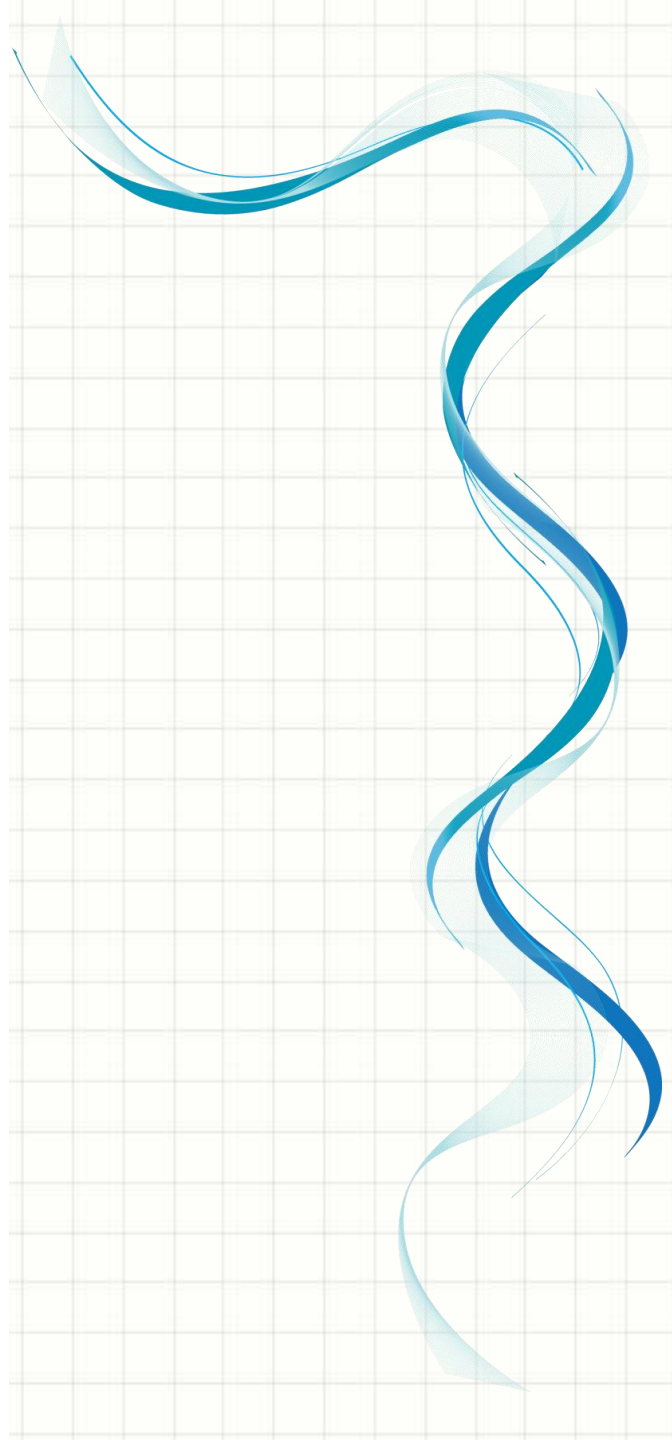
LECTURE-09

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Today's Overview:

☐ *Clippers*



A decorative blue wavy line on the left side of the slide, consisting of several overlapping, flowing curves in various shades of blue.

Clippers

Clippers:

Clippers are networks that employ diodes to “clip” away a portion of an input signal without distorting the remaining part of the applied waveform.

In general, clippers are classified into two types:

- *Series clippers* and
- *Shunt clippers*

Series clippers are again classified into two types:

- *Series negative clippers* and
- *Series positive clippers*

Shunt clippers are also classified into two types:

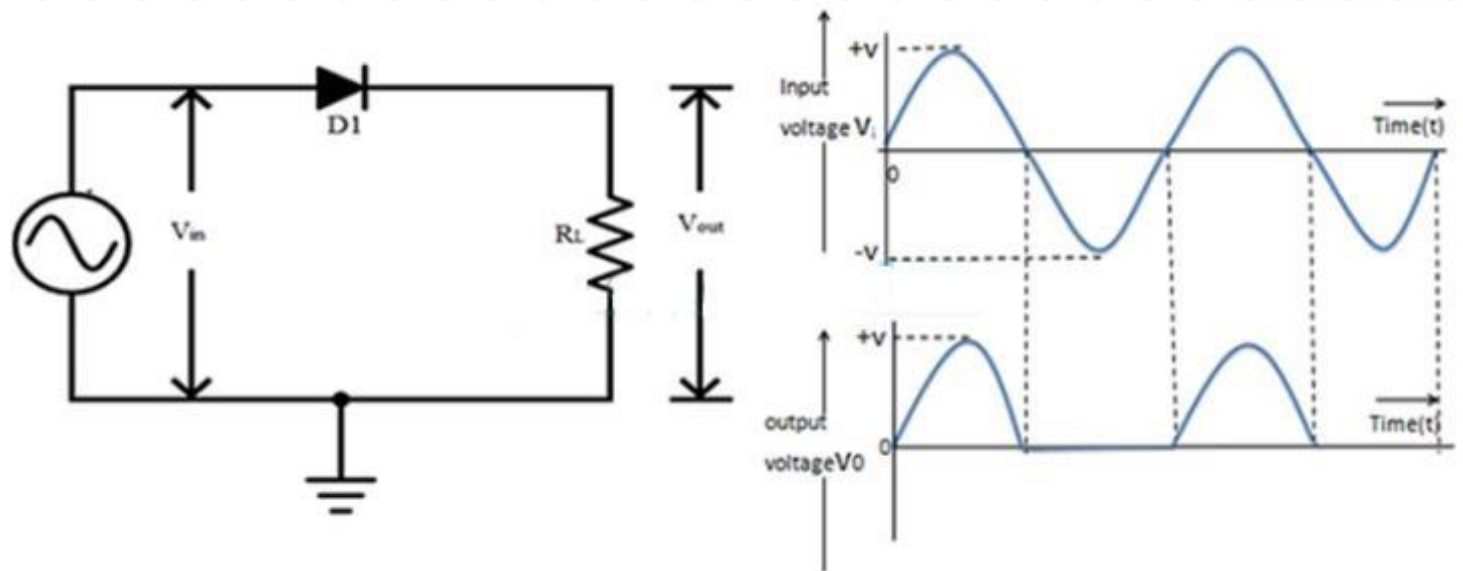
- *Shunt negative clippers* and
- *Shunt positive clippers*

Clippers:

❑ Series Negative Clipper:

During the positive half cycle, the diode (considered as an ideal diode) appears in the forward biased and conducts such that the entire positive half cycle of input appears across the resistor connected in parallel as an output waveform.

During the negative half cycle, the diode is in reverse biased. No output appears across the resistor. Thus, it clips the negative half cycle of the input waveform, and therefore, it is called a **Series negative clipper**.

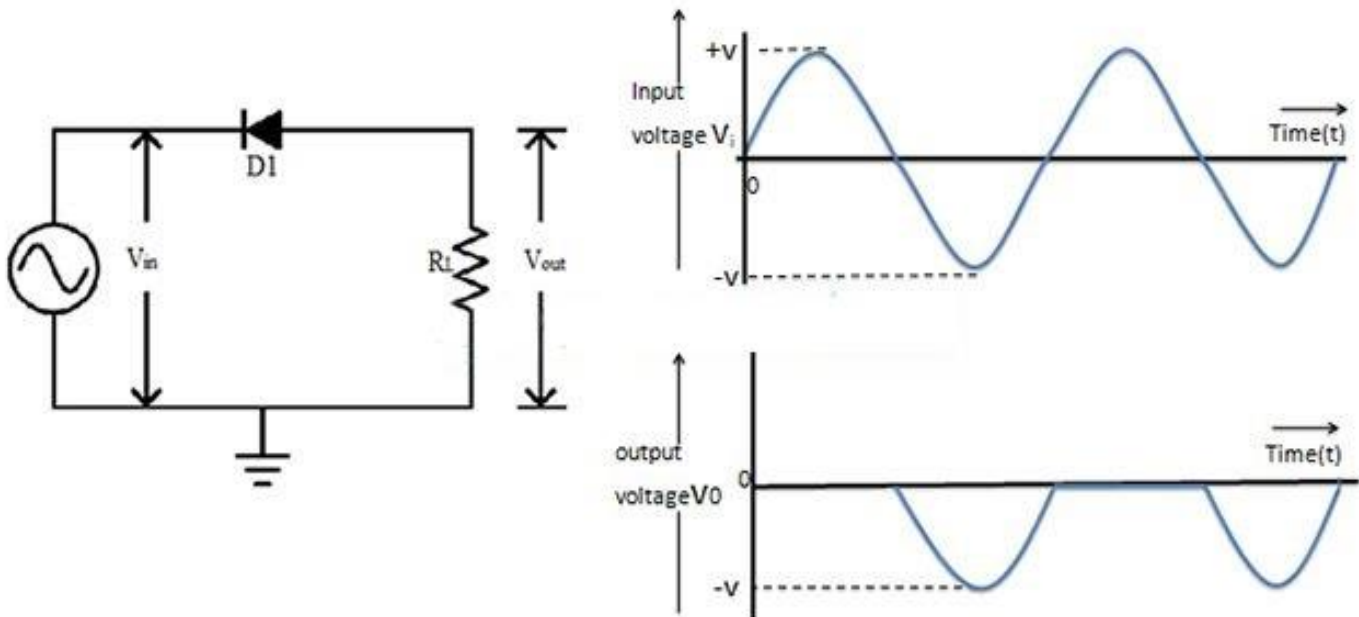


Clippers:

□ Series Positive Clipper:

The **Series positive clipper** circuit is connected as shown in the figure. During the positive half cycle, the diode becomes reverse-biased, and no output is generated across the resistor.

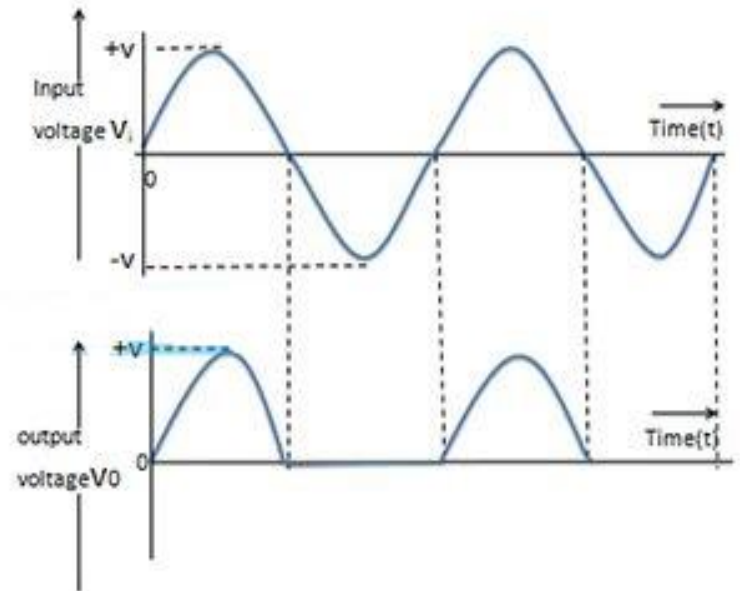
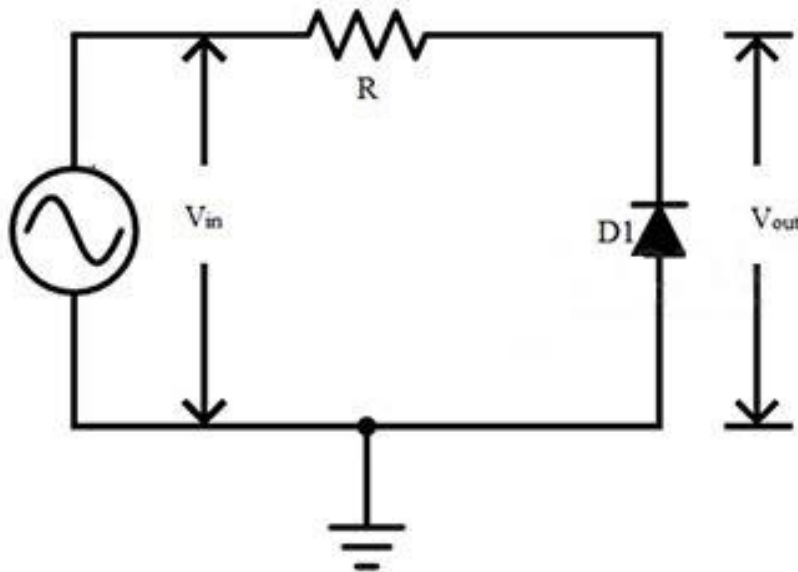
During the negative half cycle, the diode conducts and the entire input, which appears as output across the resistor.



Clippers:

□ Shunt Negative Clipper:

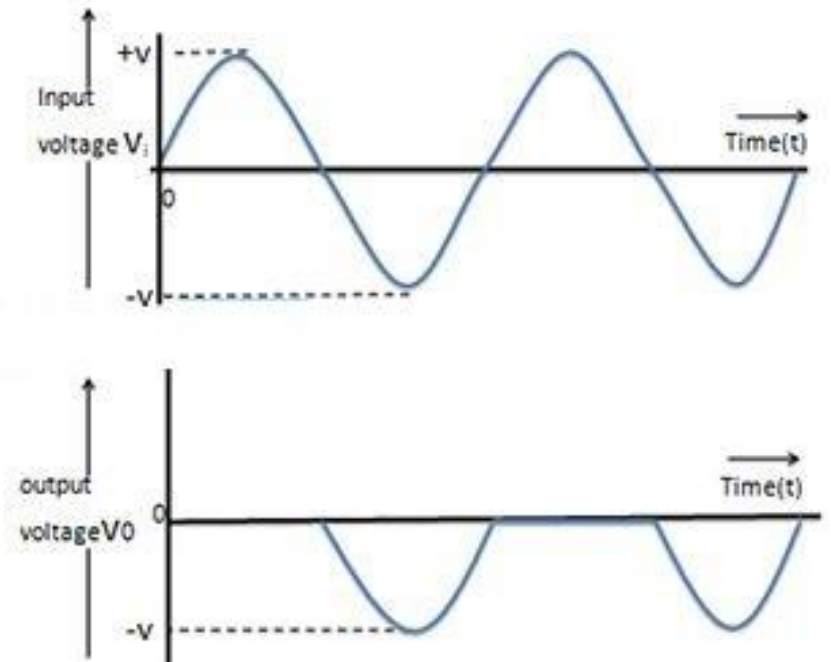
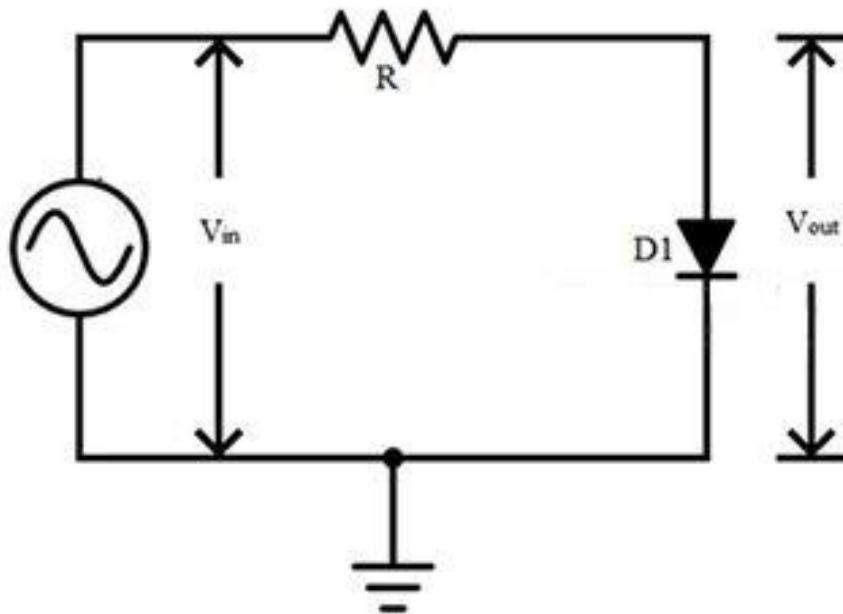
Shunt negative clipper is connected as shown in the figure. During the positive half cycle, the entire input is the output, and during the negative half cycle, the diode conducts through it, which results no output to be generated.



Clippers:

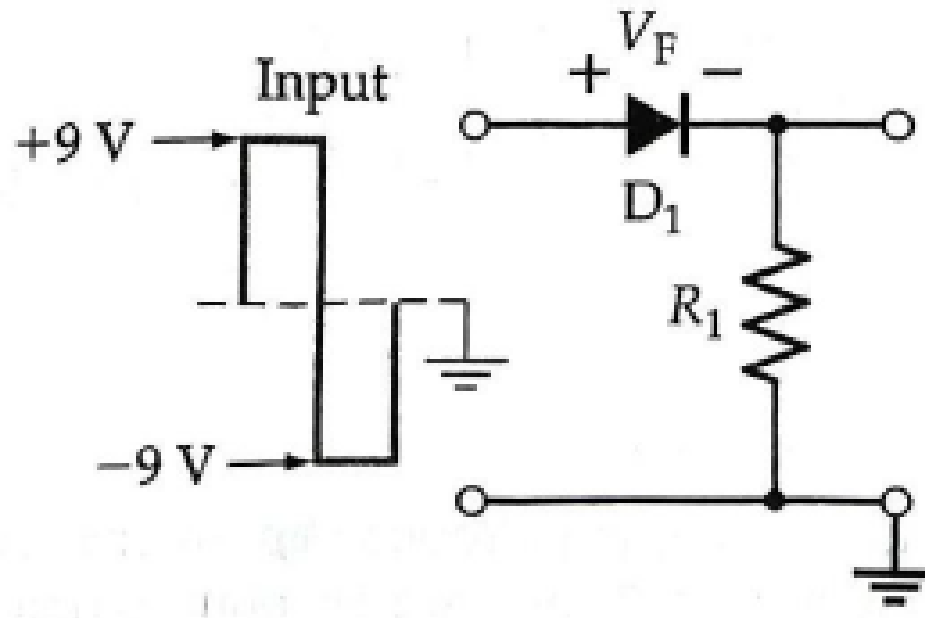
□ Shunt Positive Clipper:

Shunt positive clipper is connected as shown in the figure. During the positive half cycle, the diode is in conduction mode and no output is generated. During the negative half cycle, the entire input appears as output as the diode is in reverse bias.



Problem:

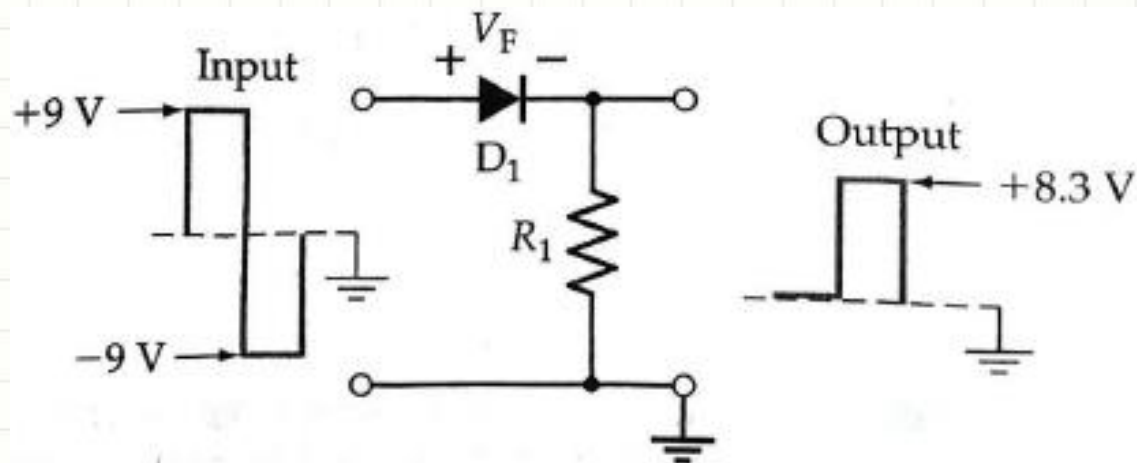
The negative series clipper in the figure has a (+/-) 9 V input and 1mA forward current. Determine a suitable resistance for R_1 , and specify the diode's output voltage. (Use a silicon diode)



Problem:

Solution

$$\begin{aligned}V_o &= E - V_F \\&= 9\text{ V} - 0.7\text{ V} \\&= 8.3\text{ V}\end{aligned}$$



Select $I_F = 1\text{ mA}$

$$\begin{aligned}R_1 &= \frac{V_o}{I_F} = \frac{8.3\text{ V}}{1\text{ mA}} \\&= 8.3\text{ k}\Omega \text{ (use } 8.2\text{ k}\Omega \text{ standard value)}\end{aligned}$$



QUESTIONS?

SEE YOU IN THE NEXT CLASS

